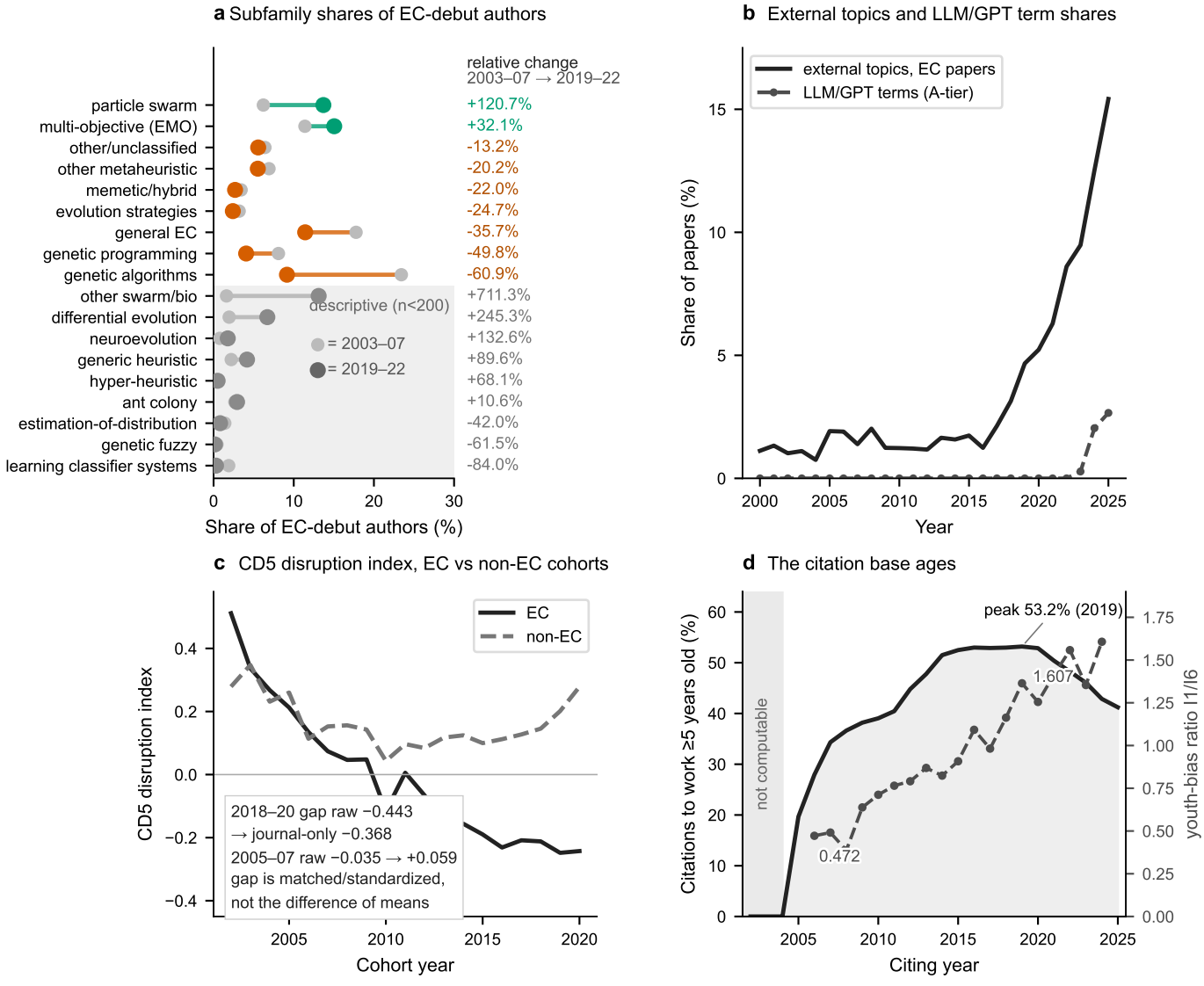
a Subfamily shares of EC-debut authors
relative change
2003–07 → 2019–22
particle swarm
multi-objective (EMO)
other/unclassified
other metaheuristic
memetic/hybrid
evolution strategies
general EC
genetic programming
genetic algorithms
other swarm/bio
differential evolution
neuroevolution
generic heuristic
hyper-heuristic
ant colony
estimation-of-distribution
genetic fuzzy
learning classifier systems
+120.7%
+32.1%
-13.2%
-20.2%
-22.0%
-24.7%
-35.7%
-49.8%
-60.9%
+711.3%
+245.3%
+132.6%
+89.6%
+68.1%
+10.6%
-42.0%
-61.5%
-84.0%
descriptive (n<200)
= 2003–07
= 2019–22
0
10
20
30
Share of EC-debut authors (%)
b External topics and LLM/GPT term shares
external topics, EC papers
LLM/GPT terms (A-tier)
Share of papers (%)
0
5
10
15
2000
2005
2010
2015
2020
2025
Year
c CD5 disruption index, EC vs non-EC cohorts
EC
non-EC
CD5 disruption index
0.4
0.2
0.0
−0.2
−0.4
2018–20 gap raw −0.443
→ journal-only −0.368
2005–07 raw −0.035 → +0.059
gap is matched/standardized,
not the difference of means
2005
2010
2015
2020
Cohort year
d The citation base ages
peak 53.2% (2019)
1.607
0.472
not computable
Citations to work ≥5 years old (%)
0
10
20
30
40
50
60
youth-bias ratio I1/I6
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2005
2010
2015
2020
2025
Citing year